

MES GROUP AR AUTOMATION SYSTEM

Revised Proposal



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Date: June 18, 2026

Version: 2.0

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1. Overview

MES AR Automation is an upload-driven platform built around MES Group's existing dormitory accounts-receivable process. It removes the repetitive work from the CSD team's desk: parsing the GIRO report, maintaining aging reports, sending the right reminders, supporting follow-up calls, and providing a clean, exportable audit trail for every action.

The system follows the billing calendar MES already runs: first SOA on the 15th, second on the 30th, email reminders on the 7th and 21st, follow-up calls mid-month (14th/15th), late-fee assessment on the 16th, and the recurring reports to CSD, management and the Relationship Managers. What changes is that each step is prepared automatically, and every action, automated or manual, leaves a verifiable trace that can be exported back into NetSuite, exactly as the MES SOP requires.

There is no integration into MES's in-house systems. The team extracts two reports (the AR Report and the DBS Bulk Collection Report) and uploads them into the dashboard. The system handles parsing, matching, queuing, drafting and reporting, and produces clean exports for manual upload back into NetSuite. Tenant data stays entirely inside MES's control.

Item	Detail
Timeline	8 Weeks
Primary Call Workflow	Manual call-log capture form
AI Voice Agent	Optional add-on (see Section 7)
Hosting (production)	Cloud, procured by MES (Azure / Hetzner / AWS)
Payment	2 tranches (PO issuance & UAT sign-off)

2. Objective

Build the MES AR Automation system, an upload-driven platform that:

- Parses the DBS Bulk Collection Report and separates the two failure reasons (NO DDA, meaning no GIRO mandate, and REFER PAYING PARTY, meaning deduction failed), then queues the correct follow-up for each.
- Reads the uploaded AR Report, reconciles it against the parsed DBS report, and flags accounts that cross the 30-day trigger line, grouped by property, GIRO status, revenue type and Live/Terminated status.
- Drafts and sends the right reminder (email reminder on the 7th, final-notice email on the 21st) based on each tenant's profile, using configurable templates.
- Provides a prioritised calling list for mid-month (14th/15th) follow-up, with a manual call-capture form as the primary path so every call reaches the same audit trail.
- Tracks promise-to-pay commitments as structured, dated records instead of free-text notes.

- Surfaces recurring-defaulter patterns (tenants whose GIRO repeatedly fails) from the parsed reports.
- Generates recurring reports on schedule: security deposits to CSD, industry-segment analysis to management, and RM balance logs.
- Provides AI-driven predictive and prescriptive analytics once production data accumulates (see Section 7).
- Produces clean exports of all activity for manual upload back into NetSuite.

3. How Reconciliation Works (Upload › Process › Export)

Because there is no system integration, the flow is deliberately simple and keeps MES in full control of the data:

1. The CSD officer extracts the AR Report and the DBS GIRO statement and uploads both into the dashboard.
2. The system parses the DBS report, separates NO DDA from REFER PAYING PARTY, and matches each failed line against the AR Report.
3. The system builds the aging board and the collections queue, applies the reminder sequence, and prepares email drafts.
4. The officer reviews and sends reminders, then makes calls and completes the post-call capture form to update each account's status.
5. All activity (emails sent, calls logged, promise-to-pay commitments) is captured and made available as a clean export to upload back into NetSuite.

On knowing whether a tenant has paid: the system does not connect to the bank in real time. When a tenant commits to pay during a mid-month call, that promise-to-pay is logged and a short confirmation reminder is sent. Final confirmation of payment comes from the following month's DBS report: if the account no longer appears in the unsuccessful-GIRO statement, it has been cleared. This matches MES's existing monthly cycle.

4. Core Features

4.1 DBS GIRO Reconciliation (Upload-Driven)

- Upload the DBS Bulk Collection Report directly into the dashboard.
- System reads every line and shows the batch summary at a glance (for example, 23 rejected out of 147).
- Splits rejections into NO DDA and REFER PAYING PARTY.
- Matches each rejection against the uploaded AR Report.
- Routes NO DDA accounts to GIRO setup and REFER PAYING PARTY accounts to balance follow-up.
- Builds the collections follow-up queue automatically.

4.2 AR Aging Board

- Reads the AR aging data from the uploaded AR Report export.
- Tenants laid out across Current / 30 / 60 / 90 / 90+ day buckets.
- Grouped by property, by revenue type, and by Live or Terminated status.
- 30-day trigger line highlighted, since that is the point follow-up begins.

4.3 Revenue-Type Segmentation (incl. 1FM)

- The AR Report contains multiple revenue services (rental, maintenance, admin fee, security deposit, etc.).
- The system separates outstanding balances by revenue type so the officer can see what each balance relates to.
- Accounts tagged 1FM (maintenance-related) are flagged separately so they can be routed and handled on their own track.

4.4 Collections Queue

- Every account needing action in one prioritised list.
- Shows reason, balance, GIRO status, revenue type and position in the reminder sequence.
- Filter by property, by 1FM tag, by revenue type, or by aging bucket.

4.5 Reminder Drafting & Review (7th / 21st)

- Configurable email templates stored in the system, so the user can edit and save template text at any time.
- The program calls the relevant template at each trigger point (reminder on the 7th, final notice on the 21st) and embeds the account-specific details automatically.
- Draft preview, edit and send, with a review step before anything goes out.

4.6 Recurring-Defaulter Insights

- Surfaces tenants whose GIRO repeatedly fails across months, drawn from the aging report and call/outcome classifications.
- Helps CSD spot chronic-risk accounts early.

4.7 Recurring Reports (pending source-document confirmation)

- Security-deposit reconciliation for CSD (Mondays).
- Industry-segment analysis for management (first working day).
- RM outstanding-balance logs (ongoing).

Note: these reports require MES to confirm whether the data is pulled directly from the AR Report or assembled from other source documents.

4.8 Late-Payment Fee Handling (pending rule confirmation)

- Identifies overdue accounts on the 16th.
- Calculates the administration fee per the MES rule.
- Updates the SOA accordingly.

4.9 Admin Dashboard (Next.js)

A single workspace for the CSD team containing:

- Upload Centre: DBS and AR file upload with parsing, matching and batch summary.
- AR Aging Board: Current/30/60/90/90+ buckets, grouped by property, revenue type and Live/Terminated, with the 30-day trigger line highlighted.
- Collections Queue: prioritised, filterable action list (property, 1FM, revenue type, aging bucket).
- Reminder Drafting & Review: template selection, edit and send, with draft preview.
- Calling List & Call Log: mid-month calling list and manual call-capture form.
- Promise-to-Pay Tracker: dated commitments with automatic follow-up.
- Recurring-Defaulter View: chronic GIRO-failure accounts.
- Reports & Export: recurring report generation and clean exports for NetSuite upload.
- Role-Based Access & Audit Trail: CSD, RM and management roles, with an access log.

5. Mid-Month Follow-Up Calls (14th15th)

The system generates a prioritised calling list for the CSD team for accounts with unreconciled or ignored balances. The primary calling workflow is the manual call-log capture form.

5.1 Manual Call-Log Capture Form (Primary Path)

Officers call clients from their normal landline or mobile, then complete a short conversation-template form in the dashboard. Key fields (client name, amount owed, deduction-fail date, aging bucket) are pre-filled from the uploaded reports. The officer captures who was reached, what was agreed, and the next action date.

- No per-minute telephony cost; calls are placed on existing MES lines.
- Every call is logged to the same audit trail as any automated activity.
- Saved as a structured, dated record and exportable for NetSuite upload.
- Traceable by date and time for later review.

This approach was chosen to keep the monthly running cost sustainable for the pilot while still producing a complete, verifiable call record.

6. Optional Add-On: AI Voice Agent

The AI voice agent is presented as an optional add-on for MES to evaluate on a go/no-go basis. It is not part of the core scope and can be enabled later if MES decides the value justifies the running cost.

6.1 How It Would Work

Built on Vapi + Twilio, the agent would place calls from a recognised Singapore caller ID that MES makes clients aware of. It discloses itself, confirms the amount owed, conveys the reminder, captures intent, and logs any promise-to-pay.

6.2 Cost & Data Trade-Offs

Consideration	Detail
Per-minute cost	~USD \$0.15/min (pay-as-you-go across STT, LLM, TTS)
Zero Data Retention (ZDR)	~USD \$1,000/mo; no recording or transcript is retained at any layer
Standard plan (no ZDR)	Voice logs & transcripts retained ~14 days on Vapi, then exported to MES's own database
Transcript availability	Only available when data is retained; not available under full ZDR
Latency	Slightly higher than a human call; may feel less natural to some clients

Recommendation: begin the pilot with the manual call-log form. If MES later wants automated calling at scale, the voice agent can be added as a costed change request once call volumes and client acceptance are understood.

7. AI Predictive & Prescriptive Analytics

Once the system is in production and time-series data accumulates, the platform can surface AI-driven inferences on top of the operational dashboards. These analytics are developed during the project in consultation with MES. The exact dashboards and inferences are agreed jointly once enough production data exists to make them meaningful. Representative examples include:

7.1 Predictive (what is likely to happen)

- Predicting which clients or client sectors are likely to continue defaulting, based on historical GIRO-failure and payment patterns.
- Identifying customer segments that are deteriorating versus improving over time.
- Flagging accounts at rising risk of crossing into 60/90+ aging buckets before they do.

7.2 Prescriptive (what to do about it)

- Recommending which accounts to prioritise for calls versus email-only follow-up.
- Suggesting the best follow-up timing and channel for each risk segment.
- Highlighting clients whose collection performance has improved, so effort can be redirected.

Scope note: the specific analytics are confirmed during development. The pilot establishes the data foundation; the analytics layer is built on top once production data is available, to avoid premature assumptions, and above are just some tentative recommendations

8. Security, Hosting & Subscriptions

8.1 Data Security

- **Data storage:** Supabase (ISO 27001, SOC 2 Type 2, GDPR-aligned), built on PostgreSQL, with Row-Level Security enforced at the data layer.

- **Hosting & access:** Front-end and backend hosted via SSH with secured user access; PM2 and Nginx for process management and endpoint security; firewalls enabled.
- **Dashboard:** Role-based access (CSD, RM, Management), user management, and an audit log of all actions.
- **AI models:** OpenAI with Zero Data Retention enabled, so MES data is not used for model training. The generative AI is used narrowly, chiefly to personalise template-based emails, while deterministic logic (aging, matching, fee calculation) runs on formulas for predictable behaviour.
- **Automation engine:** n8n, self-hosted on the MES-owned VPS.
- **Data controls:** Encryption at rest for sensitive information, role-based access, and an access log.

Because the system is upload/export-driven, tenant data stays inside MES's control, with no live connection into in-house systems is required.

8.2 Cloud Hosting Requirements

For initial testing, the system can run on DeVinci Codes' server at no cost to MES. For production, MES procures and owns the cloud subscription, and DeVinci Codes guides the migration. MES retains ownership so that any future migration does not require returning to the vendor.

Option	Notes
Hetzner VPS	Most cost-effective; a virtual private server hosting both front-end and backend
Azure	Suitable if MES standardises on Azure
AWS	Compatible with MES's existing AWS-based platforms

8.3 Ongoing Subscriptions (Production)

Service	Purpose	Notes
VPS hosting (Hetzner/Azure/AWS)	Hosting front-end, backend, n8n	Procured & owned by MES
Supabase	Database	Subscription tier per data volume
OpenAI (ZDR)	Email personalisation / analytics	Zero data retention enabled
Vapi + Twilio	Optional voice agent only	Only if AI voice add-on is enabled

8.4 Domain & Email Hosting (provided by MES)

- The email for sending followups can be setup in n8n as a credential by MES

9. Singapore Compliance Notes

- No Singapore law requires a call to be labelled as AI-generated; the Model AI Governance Framework is voluntary and encourages transparency. Where the optional AI agent is used, it discloses itself as best practice.
- The Do-Not-Call (DNC) provisions under the PDPA cover marketing messages. Collection calls to existing tenants about a live tenancy are account servicing, not marketing, and generally fall outside the DNC rules.
- The PDPA's general data-protection obligations apply to tenant numbers, records and any transcripts, and the system is built to meet them.
- All-party recording consent is best practice and is satisfied by the agent's opening disclosure where the optional voice agent is used.

10. Technology Stack

Layer	Technology
Frontend + Backend	Next.js 14 + TypeScript
Automation Engine	n8n (open source, self-hosted)
Database	Supabase (PostgreSQL)
AI / LLM	OpenAI with Zero Data Retention
Voice Agent (optional)	Vapi
Telephony (optional)	Twilio (Singapore DID)
Hosting (pilot/test)	DeVinci server; production cloud owned by MES

11. Project Milestones & Timeline (8 Weeks)

The project runs across the seven stages below. Each milestone is built and demonstrated for MES review before moving to the next. The GUI prototype is sign-off-gated: development proper begins only after the prototype is approved.

#	Stage	Indicative Timeline
1	Requirement Design Document	Week 1
2	GUI Prototype (screens, flow, transitions), MES sign-off gate	Weeks 2-3
3	Application Development	Weeks 3-5
4	Functional Testing	Week 6
5	User Acceptance Testing (UAT)	Week 7
6	Deployment	Week 8
7	Post-Deployment Stabilisation	30 days after go-live

12. Handover Deliverables (at UAT)

The following are handed over to MES at UAT:

- Source code for the full application.
- Functional Design Documents.
- User Manual.
- Clean NetSuite export format specification for all activity.
- Deployment and configuration documentation.

13. Annual Maintenance Contract (AMC)

DeVinci Codes provides 30 days of post-go-live bug-fix support (stabilisation) included in the project cost. Beyond that, an optional Annual Maintenance Contract covers ongoing support, minor enhancements and platform upkeep.

- Bug fixes and issue resolution beyond the 30-day stabilisation window.
- Minor configuration changes (e.g. template updates, report tweaks).
- Dependency, security and platform maintenance.

AMC scope and fee to be agreed before go-live. We expect this to be a light, short-term arrangement as discussed.

14. Open Items to Confirm

- How the late-payment administration fee is calculated: flat fee or percentage, any minimum or cap, which balances it applies to, whether terminated accounts differ, and the source document it is computed from.
- Current SOA, first-reminder, final-notice and 1FM templates.
- Whether recurring reports (security deposit, industry segment, RM balance) are pulled from the AR Report or assembled from other source documents.
- The Singapore caller-ID number (only if the optional voice agent is later adopted).

15. Next Steps

6. Approve the proposal and issue the Purchase Order;
7. Provide the items needed to start: current SOA and reminder templates; representative DBS and AR exports (anonymised is fine) to validate parsing; the late-payment fee rule and its source document; and source documents for the security-deposit, industry-segment and RM reports.
8. DeVinci Codes delivers the Requirement Design Document and GUI prototype for sign-off, then proceeds through development, testing, UAT, deployment and stabilisation.

17. Appendix: Source Data Files

The following source files shared by MES form the baseline inputs the system parses and reconciles. They are referenced here for completeness and used to validate parsing during development.

17.1 AR reports-Final.xlsx

Worksheet	Contents
Detailed Full Report	Master AR aging detail: customer, transaction, dates, aging bucket, open balance
Stamp Duty	Stamp-duty revenue-type outstanding lines
Park Fee	Park-fee revenue-type outstanding lines
Late Fee	Late-fee / admin-fee outstanding lines
Industry	Industry-segment analysis by aging bucket (management report)
RM - User1	Relationship Manager outstanding-balance log (User 1)
RM - User2	Relationship Manager outstanding-balance log (User 2)
Contact Details	Company name, total and email address for reminders

17.2 Other Baseline Inputs

- DBS Bulk Collection Report (unsuccessful-GIRO statement): the second upload the system parses for NO DDA and REFER PAYING PARTY rejections.
- Accounts Receivable Process Flow (MES SOP reference).